THE COLLISION TRANSFORM

ALEXANDER S. PETTY

Abstract. The collision invariant $S_\ell(p)$, introduced in the companion paper [4], is a function of $p \bmod b^{\ell+1}$ and therefore lives on the finite group $(\mathbb{Z}/b^{\ell+1}\mathbb{Z})^\times$. Its Fourier expansion over Dirichlet characters modulo $b^{\ell+1}$ is the *collision transform*. The reflection identity forces all even-character coefficients of the centered invariant to vanish: only odd characters contribute.

The centered prime harmonic sum $F^\circ(s) = \sum_p S_p^\circ/p^s$ is therefore a finite linear combination of non-trivial odd character sums $\sum_p \chi(p)/p^s$, with no principal-character term. At $s = 1$, each sum converges by Mertens' theorem for arithmetic progressions. Convergence below $s = 1$ is conditional on the absence of L-function zeros above a given depth. Computation indicates convergence persists to at least $s = 0.6$ in base 10 and to $s = 0.5$ in base 3. The real parts of the products $\hat{S}^\circ(\chi) \cdot P(s, \chi)$ have mixed signs, so convergence is a collective constraint on the joint zero distribution, not a test of each L-function individually.

Aggregating the collision deviation across bases with a fixed convergent weighting produces the *base sum*, a function on primes that reveals mod-3 structure. For bases with $3 \nmid b$, the reflection $a \mapsto m - a$ fixes a unique residue class modulo 3, and the mean of S over units in that class equals the grand mean $-1/2$ (the neutrality theorem). Removing the mod-3 component introduces a principal-character term that is absent from F°. The base-summed harmonic sum is negligible: the collision invariant's structural content is base-specific.

1. The Transform

By the finite determination theorem [4], the collision deviation S_ℓ is a function on the finite group $(\mathbb{Z}/m\mathbb{Z})^\times$ where $m = b^{\ell+1}$.

Definition 1. The *collision transform* of S_ℓ is

$$\hat{S}_\ell(\chi) = \frac{1}{\phi(m)} \sum_{a \in (\mathbb{Z}/m\mathbb{Z})^\times} S_\ell(a)\, \overline{\chi}(a)$$

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for each Dirichlet character χ modulo m, where $S_\ell(a)$ is the class function on $(\mathbb{Z}/m\mathbb{Z})^*$ from [4, Definition 4.2]. The inverse transform recovers S_ℓ:

$$S_\ell(a) = \sum_{\chi \bmod m} \hat{S}_\ell(\chi)\,\chi(a), \qquad a \in (\mathbb{Z}/m\mathbb{Z})^*.$$

In particular, for any prime $p > m$ with $\gcd(p, b) = 1$: $S_\ell(p) = S_\ell(p \bmod m)$.

This is standard Fourier analysis on the finite abelian group $(\mathbb{Z}/m\mathbb{Z})^\times$. The novelty is the function being analyzed, not the tool.

2. Structural Properties of the Transform

The reflection identity from [4] constrains the transform in a way that has immediate consequences for which L-functions can participate.

Theorem 2 (Trivial coefficient)**.** $\hat{S}_\ell(\chi_0) = -1/2$.

Proof. By the reflection identity [4], the map $a \mapsto m - a$ pairs units with $S(a) + S(m-a) = -1$. Each pair averages to $-1/2$. □

Definition 3. For a unit $a \in (\mathbb{Z}/m\mathbb{Z})^*$ (with $m = b^{\ell+1}$), the *spectral class* is $R(a) = (a-1) \bmod b$. The spectral classes are the cosets of the subgroup $\{u : u \equiv 1 \pmod b\}$ in $(\mathbb{Z}/m\mathbb{Z})^*$.

The *centered collision deviation* is $S^\circ(a) = S(a) - \overline{S}_{R(a)}$, where $\overline{S}_R$ is the average of S ([4, Definition 4.2]) over units in spectral class R.

Lemma 4 (Class reflection)**.** *For any unit a with $R(a) = (a-1) \bmod b$: $R(m-a) = b - 2 - R(a)$.*

Proof. $(a-1) + (m-a-1) = m - 2 \equiv b - 2 \pmod b$. Since a is a unit, $a \not\equiv 0 \pmod b$, so $R(a) = (a-1) \bmod b \in \{0, \ldots, b-2\}$. Similarly $R(m-a) \in \{0, \ldots, b-2\}$. Their sum is congruent to $b-2$ modulo b and lies in $\{0, \ldots, 2b-4\}$, so it equals $b - 2$. □

Lemma 5 (Class-average identity)**.** $\overline{S}_R + \overline{S}_{b-2-R} = -1$ *for every* $R \in \{0, \ldots, b-2\}$.

Proof. The reflection $a \mapsto m - a$ is a bijection on $(\mathbb{Z}/m\mathbb{Z})^*$ that sends class R to class $b-2-R$ (Lemma 4). Since the bijection maps class R onto class $b-2-R$, the two classes have equal cardinality. By the reflection identity [4], $S(a) + S(m-a) = -1$ for every unit a. Averaging over all units in class R: each $S(a)$ in class R is paired with $S(m-a)$ in class $b-2-R$, and each pair sums to -1. Therefore $\overline{S}_R + \overline{S}_{b-2-R} = -1$. □

Theorem 6 (Antisymmetry)**.** *The centered transform satisfies* $\hat{S}^\circ(\chi) = 0$ *for every even character* ($\chi(-1) = 1$)*.*

Proof. By Lemmas 4 and 5:

$$S^\circ(a)+S^\circ(m-a) = \big[S(a)+S(m-a)\big]-\big[\overline{S}_R+\overline{S}_{b-2-R}\big] = -1-(-1) = 0.$$

For even χ ($\chi(-1) = 1$), $\overline{\chi}(m-a) = \overline{\chi}(-a) = \overline{\chi}(a)$, so

$$\hat{S}^\circ(\chi) = \frac{1}{\phi(m)} \sum_a S^\circ(a)\,\overline{\chi}(a) = -\hat{S}^\circ(\chi),$$

hence $\hat{S}^\circ(\chi) = 0$. □

Corollary 7. *The centered collision deviation decomposes as*

$$S^\circ(p) = \sum_{\substack{\chi \bmod m \\ \chi(-1)=-1}} \hat{S}^\circ(\chi)\,\chi(p).$$

Only odd characters contribute. The bilateral symmetry of the digit function determines which L-functions the collision transform can see.

All identities involving F° and $P(s,\chi) = \sum_p \chi(p)/p^s$ are understood up to the finite contribution of primes $p \le m$, which is irrelevant for convergence.

3. Convergence at $s = 1$

The question that matters is whether the centered collision deviations cancel across the population of primes.

Theorem 8 (Centered convergence)**.** *For every base* $b \ge 2$ *and lag* $\ell \ge 1$*,*

$$F^\circ(1,\ell) = \sum_{\substack{p>m \\ \gcd(p,b)=1}} \frac{S_p^\circ(\ell)}{p}$$

converges.

Proof. By the antisymmetry theorem,

$$F^\circ(1) = \sum_{\substack{\chi \bmod m \\ \chi(-1)=-1}} \hat{S}^\circ(\chi) \sum_p \frac{\chi(p)}{p}.$$

For each non-trivial character χ modulo m, the prime harmonic sum $\sum_p \chi(p)/p$ converges. This follows from Mertens' theorem for arithmetic

progressions [3, 1]: the prime sum in each residue class modulo m satisfies

$$\sum_{\substack{p\le x\\ p\equiv a \pmod m}} \frac{1}{p} = \frac{\log\log x}{\phi(m)} + M_a + o(1),$$

and the character weighting $\sum_a \chi(a) = 0$ cancels the $\log\log x$ term. Since F° is a finite linear combination of convergent sums, it converges. □

4. Below $s = 1$

Convergence at $s = 1$ asks little of each prime: the weight $1/p$ decays rapidly. Lowering s amplifies every prime's contribution, demanding that finer structure project coherently across the population.

For $\mathrm{Re}(s) > 1$, the prime character sum $P(s,\chi) = \sum_p \chi(p)/p^s$ relates to the Dirichlet L-function [1, 2] by

$$P(s,\chi) = \log L(s,\chi) - H(s,\chi),$$

where $H(s,\chi)$ converges for $\mathrm{Re}(s) > 1/2$.

Theorem 9 (Conditional penetration)**.** *Suppose there exists $\sigma_0 \in (1/2, 1)$ such that for every odd character χ modulo m, the L-function $L(s,\chi)$ has no zeros in the half-plane $\mathrm{Re}(s) > \sigma_0$. Then $F^\circ(s)$ converges for every real $s > \sigma_0$.*

Proof. By the antisymmetry theorem, $F^\circ(s) = \sum \hat{S}^\circ(\chi)\, P(s,\chi)$ over odd non-trivial characters. It suffices to prove $P(s,\chi)$ converges for each such χ and real $s > \sigma_0$.

First assume χ is primitive. The explicit formula for $\psi(x,\chi) = \sum_{n\le x} \Lambda(n)\chi(n)$ gives [1, 2]

$$\psi(x,\chi) = -\sum_{|\gamma|\le T} \frac{x^\rho}{\rho} + O\left(\frac{x\log^2(mxT)}{T}\right),$$

where $\rho = \beta + i\gamma$ runs over non-trivial zeros of $L(s,\chi)$. By hypothesis $\beta \le \sigma_0$. Choosing $T = x$ and applying the standard zero-sum estimate (see [1], Ch. 17, or [2], Theorem 5.15):

$$\psi(x,\chi) \ll x^{\sigma_0}\log^2(mx).$$

Since $\psi - \vartheta \ll x^{1/2}$ and $\sigma_0 > 1/2$, we have $\vartheta(x,\chi) \ll x^{\sigma_0}\log^2(mx)$, where $\vartheta(x,\chi) = \sum_{p\le x}\chi(p)\log p$.

Partial summation with $f(t) = 1/(t^s \log t)$ gives

$$\sum_{p\le X} \frac{\chi(p)}{p^s} = \frac{\vartheta(X,\chi)}{X^s\log X} + \int_2^X \vartheta(t,\chi)\,\frac{s\log t + 1}{t^{s+1}(\log t)^2}\,dt.$$

The boundary term tends to 0, and the integral converges because $s > \sigma_0$. Thus $P(s,\chi)$ converges. If χ is imprimitive, induced from a primitive χ^*, then $P(s,\chi) - P(s,\chi^*)$ is a finite sum over primes dividing m, so convergence is unchanged. Thus F° converges as a finite linear combination. □

Proposition 10 (Analytic continuation)**.** *Define*

$$\mathcal{F}^\circ(s) = \sum_{\chi \text{ odd}} \hat{S}^\circ(\chi)\big(\log L(s,\chi) - H(s,\chi)\big).$$

Then $\mathcal{F}^\circ$ agrees with F° for $\mathrm{Re}(s) > 1$ *(up to the finite contribution of primes $p \le m$) and extends holomorphically to every simply connected region in* $\mathrm{Re}(s) > 1/2$ *on which all relevant odd $L(s,\chi)$ are nonvanishing.*

Proof. For $\mathrm{Re}(s) > 1$, the Euler product gives $P(s,\chi) = \log L(s,\chi) - H(s,\chi)$, so $\mathcal{F}^\circ(s) = F^\circ(s)$. On a simply connected zero-free region Ω in $\mathrm{Re}(s) > 1/2$, each $L(s,\chi)$ admits a holomorphic branch of $\log L(s,\chi)$, and $H(s,\chi)$ is holomorphic on Ω. The finite sum $\mathcal{F}^\circ$ is therefore holomorphic on Ω. □

s	base 3	base 7	base 10	base 12
1.0	−0.15	−0.14	−0.20	−0.28
0.8	−0.19	−0.25	−0.40	−0.62
0.6	−0.23	−0.46	−0.84	−1.35
0.5	−0.26	−0.64	−1.22	−1.88

TABLE 1. $F^\circ(s)$ across bases at 348,488 primes [5]. Computation indicates convergence persists well below $s = 1$.

Remark 11. The products $\mathrm{Re}(\hat{S}^\circ(\chi) \cdot P(s,\chi))$ have mixed signs: in base 10 at $s = 0.5$, thirteen of twenty are positive and seven are negative. The collision coefficients $|\hat{S}^\circ(\chi)|$ are negatively correlated with $|P(s,\chi)|$ (Pearson $\rho \approx -0.25$ across all prime bases tested). The collision invariant appears to place larger Fourier weight on characters less sensitive to L-function zeros. These observations suggest that any persistence of convergence below $s = 1$ would be a collective phenomenon, constraining the joint zero distribution of the odd L-functions modulo m rather than each L-function individually.

5. The Base Sum

For a fixed prime p, the collision deviation $S_1(p, b)$ varies across bases. The *base sum*

$$R(p) = \sum_b w(b)\, S_1^\circ(p, b),$$

with a fixed convergent weighting $w(b)$, aggregates collision structure across bases. Since S° depends on $p \bmod b^2$ at each base b, the function $R(p)$ combines information from the finite groups $(\mathbb{Z}/b^2\mathbb{Z})^\times$ across all bases.

Computation reveals that $R(p)$ has mod-3 structure: at lag 1 in every base b with $3 \nmid b$, the mean of S over units $a \equiv 2 \pmod 3$ equals exactly $-1/2$ (the grand mean), while the mean over $a \equiv 1 \pmod 3$ deviates. This observation, stable across all bases tested, motivates the neutrality theorem in the following section.

Observation 12. The base-summed prime harmonic sum

$$F_R(s) = \sum_p \frac{R(p)}{p^s} = \sum_b w(b)\, F_b^\circ(s)$$

nearly vanishes. At $s = 0.5$ with $w(b) = 1/b^2$ and bases $b = 3, \ldots, 31$: individual $F_b^\circ(0.5)$ values range from -1.4 to $+5.1$, but their weighted sum is -0.01. The collision invariant's structural content appears to be base-specific.

6. The Mod-3 Structure

Theorem 13 (Neutrality)**.** *Let $3 \nmid b$ and $m = b^{\ell+1}$. Let k^* be the unique solution to $2k \equiv m \pmod 3$ (i.e., $k^* = 2$ if $m \equiv 1$ and $k^* = 1$ if $m \equiv 2 \pmod 3$). Then the mean of S_ℓ over units $a \equiv k^* \pmod 3$ in $(\mathbb{Z}/m\mathbb{Z})^\times$ equals $-1/2$ (the grand mean).*

Proof. The reflection $a \mapsto m - a$ sends class k to class $m - k \pmod 3$. It fixes the class k^* satisfying $k^* \equiv m - k^* \pmod 3$. Since 3 is odd, the solution is unique. The reflection identity $S(a) + S(m-a) = -1$ [4] pairs units within this class, and each pair averages to $-1/2$. □

Corollary 14. *The remaining two classes modulo 3 (other than k^*) are swapped by reflection, and their means satisfy $\overline{S}_k + \overline{S}_{m-k} = -1$.*

Remark 15. At lag $\ell = 1$, $m = b^2 \equiv 1 \pmod 3$ for every b with $3 \nmid b$ (since $1^2 \equiv 2^2 \equiv 1$). In this case $k^* = 2$: class 2 is neutral and class 1 carries the bias. Since primes $p > 3$ satisfy $p \bmod 3 \in \{1, 2\}$, the prime harmonic sum sees class 1 (biased) and class 2 (neutral).

Theorem 16 (Perfect cancellation)**.** *The centered sum F° contains no Mertens term. Its decomposition over characters modulo m involves only non-trivial characters:*

$$F^\circ(s) = \sum_{\substack{\chi \bmod m \\ \chi(-1)=-1}} \hat{S}^\circ(\chi)\, P(s,\chi).$$

Proof. By the antisymmetry theorem, only odd characters contribute. Every odd character is non-trivial (the trivial character is even). Therefore no term $\sum_p p^{-s}$ appears. □

Corollary 17. *Let $\mu_3(a)$ be the mean of S° over all units $a' \equiv a \pmod 3$ in $(\mathbb{Z}/m\mathbb{Z})^\times$. For primes $p > 3$, write $\mu_3(p) = c_0 + c_1\chi_3(p)$, where χ_3 is the non-trivial character modulo 3 and $c_0 = (\mu_3^{(1)} + \mu_3^{(2)})/2$ is the average of the two mod-3 class means visible to primes. Set $\mathcal{M}(s) = \sum_p \mu_3(p)/p^s$ and $F^{\circ\circ}(s) = F^\circ(s) - \mathcal{M}(s)$.*

When $c_0 \neq 0$, the sum $\mathcal{M}$ contains a principal-character term $c_0 \sum_p p^{-s}$, and $F^{\circ\circ}$ carries $-c_0 \sum_p p^{-s}$ to compensate. These principal-character terms cancel algebraically in F° by Theorem 16.

s	F°	$F^{\circ\circ}$
1.0	0.08	0.77
0.7	0.27	12.3
0.5	0.46	111

Table 2. Numerically, removing the mod-3 component destroys convergence below $s = 1$. Base 10, 348,488 primes.

Remark 18. In base 3 at lag 1, $m = 9 \equiv 0 \pmod 3$, so the neutrality theorem does not apply (all units satisfy $a \not\equiv 0$). At lag 1 in bases with $3 \nmid b$, $m = b^2 \equiv 1 \pmod 3$ and class 2 is neutral. In general, the neutral class k^* depends on m mod 3.

7. Remarks

The collision transform converts a digit-dynamic invariant into a standard object of analytic number theory. The invariant is new; the tools applied to it are classical. Convergence at $s = 1$ is a theorem. Convergence below $s = 1$ is conditional on the absence of L-function zeros and is controlled by the sign structure of the character decomposition.

The antisymmetry theorem selects which L-functions participate (odd characters only). The neutrality theorem and perfect cancellation eliminate the principal-character term. The observed anti-correlation between collision coefficients and prime character sum magnitudes is consistent with the persistence of convergence: the collision invariant's Fourier spectrum appears to be misaligned with the zero-sensitive directions in character space.

References

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Email address: `alexander.petty@gmail.com`